

Isuru Rajapakshe

Los Angeles, CA

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Summary

Electrical Engineer specializing in sensing systems, signal processing, and embedded development. Experience spanning LiDAR perception systems, neuroscience research, experimental hardware validation, and real-time data processing. Strong background in Python-based analysis, machine learning, and sensor characterization for complex engineering systems.

Technical Skills

Programming: Python, C, JavaScript, Shell Scripting

Signal Processing & Analysis: FFTs, Spectral Analysis, Filtering, Sensor Characterization, EEG Analysis

Embedded Systems: Zephyr RTOS, nRF52, PIC Microcontrollers, Arduino, Seeed XIAO

Machine Learning: PyTorch, MATLAB

Data & Visualization: NumPy, Pandas, Plotly, Open3D

Engineering Tools: Git, SolidWorks, OrCAD, Proteus, Firebase

Experience

Development Engineer

Sep 2025 – Present

SiLC Technologies

Los Angeles, CA

- Develop and validate FMCW LiDAR systems for perception and ranging applications.
- Build Python pipelines for point cloud processing, range precision analysis, and sensor characterization.
- Conduct testing and validation across optical, electrical, and embedded subsystems.
- Develop tools for automated visualization and performance analysis.

Course Instructor / Instructional Assistant

Jun 2024 – Dec 2025

California State University Los Angeles

Los Angeles, CA

- Taught Introduction to Biomedical Devices, Digital Logic Laboratory, and Biomedical Engineering Laboratory courses.
- Guided interdisciplinary student teams through hardware-software integration and project development.

Software Engineer

Jul 2021 – Jul 2022

Insync Information Technologies

Colombo, Sri Lanka

- Led development of a network automation platform using the MongoDB-Express-React-NodeJS stack.
- Designed software workflows to automate network configuration, monitoring, and management tasks.

Electronic Engineering Intern

Jun 2019 – Dec 2019

Lanka Electronics

Colombo, Sri Lanka

- Developed trajectory planning and welding control features for a 6-DOF robotic arm in C#.
- Improved robotic positioning accuracy to approximately 3 mm.

Projects

EEG Biomarker Analysis for rTMS Therapy

Dec 2023 – May 2025

Research Assistant, California State University, Los Angeles

- Developed MATLAB and Python pipelines for EEG preprocessing, spectral analysis, and automated feature extraction.
- Characterized EEG biomarkers associated with rTMS treatment response in Major Depressive Disorder.
- Applied signal processing and statistical techniques to analyze multi-channel neural datasets.

Gesture-Controlled Music Generation System

Jun 2024 – Aug 2024

Student Mentor, BOOST Program, California State University, Los Angeles

- Led a five-member team in developing a gesture-controlled music generation device using real-time hand tracking.
- Delivered a rehabilitation system designed to support motor recovery in stroke survivors at the Cal State LA Mobility Center.

LipRead – Visual Speech Recognition Pipeline

Nov 2023 – Dec 2023

Graduate Course Project, California State University, Los Angeles

- Developed a PyTorch-based visual speech recognition pipeline using GRUs.
- Achieved 66% word prediction accuracy on the evaluation dataset.

Non-Intrusive Load Monitoring (NILM) System

Feb 2020 – Jul 2021

Undergraduate Capstone Project, University of Moratuwa

- Developed an appliance-level energy monitoring system using embedded sensing and machine learning.
- Designed power metering circuitry in OrCAD and developed Firebase-based monitoring tools.
- Achieved 25% reduction in mean absolute error compared to state of the art methods

Education

Master of Science in Electrical Engineering — GPA: 4.0/4.0

Aug 2023 – May 2025

California State University, Los Angeles

Los Angeles, CA

Bachelor of Science in Engineering (Hons.) — GPA: 3.57/4.2

Oct 2016 – Jun 2021

University of Moratuwa

Sri Lanka